# NetworkPolicies

Restricting In-cluster Network Traffic

# Need

By default, all network traffic within a cluster is allowed

When a **NetworkPolicy** is a applied to a pod, change to *default-deny*
→ Traffic is blocked unless specifically allowed

Helps block:

- Badly-behaved workloads within the cluster
- Attacker who gain access to a pod in the cluster

# Creating a Network Policy

```yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: policyname
  namespace: ns     # Applies to pods in this namespace
spec:
  podSelector: {}  # If empty, applies to all pods in namespace.
  policyTypes:     # Otherwise, select which pods get this policy.
  - Ingress   # Can restrict ingress (traffic into the pod),
  - Egress    # egress (traffic out of the pod), or both
  # ...
```

# Ingress Rules

```yaml
spec
  ingress:
  - from:    # A single rule to allow traffic
    - namespaceSelector:
        matchLabels:
          kubernetes.io/metadata.name: my-ns  # Allow traffic only from
    ports:                                    # from pods in my-ns
    - protocol: TCP  # Allow only TCP traffic (as opposed to UDP)
      port: 2000     # Allow traffic only on ports 2000-2004
      endPort: 2004  # endPort can be omitted for only a single port
```

All other traffic into these pods will be blocked

# Egress Rules

```
egress:
  - ports:                # First rule:
    - port: 53            # Allow outbound traffic to anywhere on port 53
      protocol: UDP       # (This generally needed for DNS)
    - port: 53
      protocol: TCP
  - to:                     # Second rule
    - podSelector:          # Allow outbound traffic to ...
        matchLabels:
          app: app-name     # ... pods with the label app=app-name ...
    ports:
      - port: 12345         # ... on port 12345.
        protocol: TCP
```